

PHYS 121 — HW4 (Redesigned): Rotation, Rolling & Impacts

Student Name: _____.

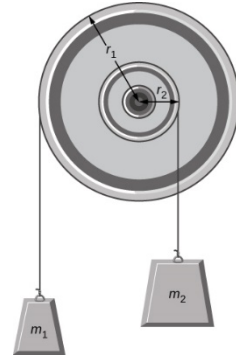
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Q1. A rotating grinding wheel (10 pts)

A uniform cylindrical grinding wheel ($M=50.0$ kg, diameter 1.0 m) is driven by a motor. Friction in bearings is negligible. (a) Spin-up torque: From rest to 120 rev/min in 20 revolutions. Draw $\omega-t$ and $\theta-t$ sketches; use rotational kinematics to find α and then $\tau=I\alpha$. (b) Steady-speed torque: A tool ($\mu_k=0.60$) is pressed perpendicularly at the rim with 40.0 N. Draw an FBD at contact; compute the friction torque and the new motor torque needed to hold constant ω . Required drawing: Top view with radius vector, applied force, friction, and torque sense arrows.

Q2. Two-groove Atwood pulley (10 pts)

A rigid pulley (moment of inertia I) has two radii r_1 and r_2 with light strings and masses m_1 and m_2 , as shown. (a) Angular acceleration: Choose positive rotation so that the heavier side (m_2) moves down. Write $\tau_{\text{net}} = I\alpha$ from the two tensions; assume no-slip. Solve for α . (b) Linear accelerations: Report $a_1 = \alpha r_1$ and $a_2 = \alpha r_2$ with directions. Required drawing: Side view with signs, radii, tensions, and acceleration arrows.



Q3. Rotating stick with end masses (10 pts)

A stick ($L=1.0$ m, $M=6.0$ kg) pivots about its center. Masses of 4.0 kg and 2.5 kg are attached to the two ends, respectively. Released from horizontal. (a) Drawing: Show pivot at center, heights of the ends initially (horizontal) and at vertical. (b) Energy method: Only the two end masses change gravitational energy. Compute ΔU . (c) Find the rotational inertia of the 'stick+masses' and compute ω at vertical. (d) Check: If $m_1=m_2$, does $\omega \rightarrow 0$? If one mass $\rightarrow 0$, does the result match a single point mass at $L/2$?

Q4. Spring-pushed sphere that skids then rolls (17 pts)

A solid ball of mass m and radius r is originally at rest on a level surface. There is friction. An elastic spring with spring constant k is compressed a length x . The spring is released onto the ball (pushing, not attaching), transferring the potential energy stored in the spring into the ball. (a) Write an expression for the translational velocity v_i of the center of mass of the ball right after it leaves the spring. At this point, assume that the ball is not rotating, and that friction with the surface is zero.

The ball then begins to skid on the surface, which starts to rotate. The ball will skid until it rotates fast enough to roll without slipping; the translational velocity of the center of mass of the ball is necessarily slowing down while this happens. The conserved quantity in this case is the angular momentum of the ball about the point of contact with the surface.

(b) When the ball is not rotating, the angular momentum is simply $L = mv_i r$. Write this quantity in terms of m , r , k , and x . (c) When the ball rotates without slipping, the angular momentum is given by $L = I\omega$. In this case, I is the moment of inertia about the point of contact. Find I in terms of m and r . You will need to use the parallel axis theorem. (d) The angular velocity ω is given by v_f/r , where v_f is the final speed of the ball once it rolls without slipping. Use conservation of angular momentum to find v_f in terms of v_i . (e) Find the percentage of initial energy stored in the spring that dissipated by friction in this process. Notice that your answer does not depend on the coefficient of friction, only that friction exists! (f) A toy consists of a ball about the size of a grape and a spring that requires a force of about 10 N to compress it about 1 cm. Estimate the final speed of the ball. (2+2+3+4+3+3 points)

Q5. Paper roll on an accelerating truck (10 pts)

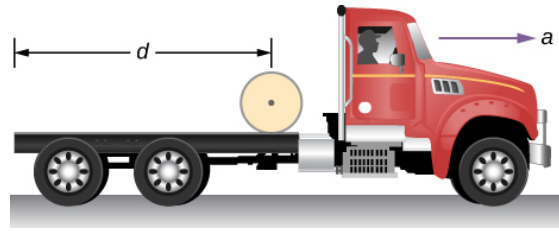
A solid cylindrical roll rests on a flat truck bed.

The truck accelerates forward at a constant a .

How far does the truck move before the roll reaches the back edge? 1) Choose frames: In the ground frame, write Newton's 2nd law for translation and rotation, respectively; (signs!).

2) No-slip on the bed: acceleration of the

contact point equals the truck's a : ($a_C + R\alpha = a$). Solve for a_C . 3) Relative motion: The roll's backward acceleration relative to the truck is ($a_{rel} = a_C - a$). 4) Distance to drop: find s , the truck's distance.



Q6. Bug hits a hanging stick (10 pts)

A bug (speed u) hits and sticks to the end of a uniform stick of length L , pivoted at its top and hanging vertically. The stick's mass is 10 times the bug's mass. After impact, the stick + bug swings to a maximum angle (θ) from vertical before returning. 1) Impact (angular momentum about pivot): find the angular speed of the 'bug+stick' right after the impact. 2) Solve for L in terms of u and θ . 3) Check: If θ goes to 0, does the required L blow up appropriately?

Q7. Ice-skating twirl and the double Axel (15 pts)

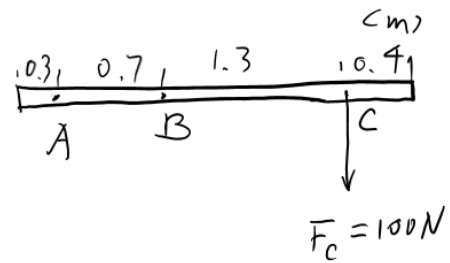
An ice skater can be thought of as a solid cylinder with a height of 1.50 meters, a radius of 10.0 cm, and a mass of 45 kg. The skater also has two arms, each treated as a uniform thin rod of length 0.60 meters and a mass of 2.4 kg that is attached to the outside of the cylinder; these thin rods can either be extended outward perpendicular to the cylinder, or held against the surface of the cylinder. We want to use our knowledge of physics to understand the double Axel, which is when an ice skater jumps up and rotates 2.5 times while in the air.

- (a) Determine the rotational inertia of the ice skater about the axis of the cylinder with
 - i. the arms at the side, and
 - ii. the arms extended.
- (b) The ice skater jumps upward to a maximum height of 0.9 meters. What length of time is the ice skater in the air?
- (c) What is the necessary rotational speed of the ice skater if she is to complete a double Axel?
- (d) The ice skater achieves this rotational speed by kicking off from the ground with arms in one configuration, and then quickly moving arms to another configuration. Which configuration is used when, and what is the initial angular speed of the jumper before changing configurations?
- (e) The skater will often land with one leg extended. Why? One sentence is enough.

(4+3+3+3+2 points)

Q8. Three-force balance on a massless rod (10 pts)

A long, massless straight rod is acted on by three perpendicular forces at labeled points A, B, C (see figure). The rod is in static equilibrium. determine the magnitude and direction of the other two forces.



Q9. Spider silk realism + Spiderman (16 pts)

The elastic modulus of silk is about $Y = 25 \text{ GPa}$ ($2.5 \times 10^{10} \text{ N/m}^2$). This means that a silk thread of length L and cross-sectional area A , then the tension in a silk thread stretched by an amount x would be $F = Y A x/L$. Consider a 50 mg spider that has a silk line with a diameter of $d = 0.14 \text{ mm}$. Assume the spider is hanging from a line that is $L = 50 \text{ cm}$ long.

- (a) By how much does the line stretch because of the weight of the spider?
- (b) How much potential energy is stored in the stretched silk line?
- (c) If all of this energy is released to the spider in the form of kinetic energy, what would be the speed of the spider?

In one of the Spiderman movies, Spiderman stops a falling car with a single silk line, although it is much thicker than the spider line above. Assume that the car falls out of a ten-story tall parking garage, the line is attached to the top of the parking garage, the silk line manages to connect to the car when it is only two stories above the ground, and the car comes to a rest right when it reaches the ground (similar to the vine jumpers in Vanuatu). I'm getting this info by watching a movie; the numbers are all approximate. (by Prof. Paul Stanley)

- (d) Draw a picture!
- (e) Estimate the minimum necessary diameter of the silk line that Spiderman must be using for the physics to work. At this point, assume the silk is infinitely elastic.
- (f) Would the line break? How do you know? At this point, don't assume the silk is infinitely elastic.

(3+3+3+1+3+3 points)

Bonus (0–2 pts): Measure a moment of inertia

Search and briefly summarize how to use a torsion(-like) pendulum to measure the moment of inertia of an irregular object. Draw a sketch to illustrate the experimental setup. [less than 200 words]

Assistance notes: